

KUAN-FU FENG

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GitHub [click] | LinkedIn Profile [click] | Google Scholar [click]

Research Seismologist with 10+ years of specialized experience in time series analysis, signal processing, and monitoring. Proficient in scientific computing, programming, cloud computing, and open-source tools. Capable of translating models into practical workflows and dedicated to applying research findings to engineering solutions.

RESEARCH PROJECT

Ambient Noise Seismology and Subsurface Monitoring

- Analyzed ambient noise to detect subtle subsurface velocity changes linked to tectonic and environmental factors.
- Implemented physical-based and statistical methods to examine environmental effects on multiscale (crust/critical zones).
- Designed noise-based seismic monitoring systems to track crustal changes linked to tectonic processes.
- Collaborated with interdisciplinary teams to enhance monitoring frameworks and data interpretation.

Time-dependent Seismic Tomography and Velocity Monitoring

- Inverted regional earthquake data to image crustal seismic velocity structures.
- Examined co-seismic velocity changes associated with moderate to large earthquakes

SKILL

Software Engineering	Linux/Unix, Git/GitHub, Jupyter Notebook, Visual Studio Code
Programming Languages	Python, Fortran, C, C++, Bash Script
Scientific/Technical Skills	Time series analysis, Signal processing, Inversion methods, Statistical analysis
Data visualization	ParaView, Matplotlib, Seaborn, Adobe Illustrator
Cloud/HPC Environment	AWS (S3, EC2, Jupyter), parallel scripting, workflow automation

PROFESSIONAL EXPERIENCE

Postdoctoral Research Associate

University of Utah, USA

Dec 2024 - present

- Applied time series analysis to geophysical datasets for subsurface monitoring, integrating multi-scale geophysical datasets.
- Integrating inversion workflows with environmental datasets to improve hydrogeophysical model accuracy.

Postdoctoral Scholar

University of Washington, USA

Jun 2023 - Nov 2024

- Developed data-driven models linking near-surface seismic changes to groundwater-surface interactions.
- Contributed code and testing to an open-source environmental seismology toolkit, improving functionality and performance.
- Assisted in delivering practical workshops on cloud computing, promoting scalable seismic data processing using AWS.

Postdoctoral Scholar

University of Utah, USA

Fellowship supported

Jun 2022 - May 2023

- Processed and analyzed over 10 TB of high-resolution ambient noise data for time-lapse velocity monitoring.
- Investigated the influence of ambient noise sources on seismic array performance at multiple scales.

Graduate Research Assistant (*PhD student*)

National Taiwan University and Academia Sinica, Taiwan

Sep 2017 - Jan 2022

- Performed time-frequency domain analysis on multi-year seismic datasets to track crustal property changes.
- Designed and implemented noise-based seismic monitoring systems for tectonic and hydrological studies.
- Analyzed earthquake sequences, rupture directivity, focal mechanisms, and velocity models to study seismogenic structures.
- Assisted with the deployment of portable and permanent seismic arrays in landslide-prone regions of Taiwan.

Research Assistant

Institute of Earth Sciences, Academia Sinica, Taiwan

Oct 2016 - Aug 2017

- Maintained and tested Real-time Earthquake Moment Tensor Monitoring System.
- Constructed finite-fault models of subduction zone events using broadband waveform modeling.

EDUCATION

PhD	Geosciences, National Taiwan University, Taiwan	Sep 2017 - Jan 2022
Master	Geosciences, National Taiwan University, Taiwan	Sep 2014 - Jun 2016
Bachelor	Earth and Environmental Sciences, National Chung Cheng University, Taiwan	Sep 2010 - Jan 2014

LEADERSHIP & SERVICE

Peer-reviewer

Reviewed manuscripts for *Geophysical Research Letters* (2), *Journal of Geophysical Research* (5), *Nature Communications* (1), *Earth, Planets and Space* (1), *Tectonophysics* (1).

Workshop instructor

Earth and Space Sciences, University of Washington

2024

- Delivered workshop on Ambient Noise Seismology in the Cloud [SCOPED Workshop (link)].
- Instructed Data Mining on the Cloud 101 at the Seismological Society of America Meeting [SSA Tutorials (link)].

President, Graduate Student Association

Department of Geosciences, National Taiwan University

2018

- Elected by peers to represent graduate students; served as a liaison between the department and the student body.
- Organized student forums and summer lectures for the department and institute to promote interdisciplinary engagement and discourse.

Summer Student Lecture Convenor and Instructor

Institute of Earth Sciences, Academia Sinica, Taiwan

Summer 2018 and 2019

- Managed logistics for an institute's summer lecture series aimed at undergraduate and early-career students.
- Designed and delivered instruction on earthquake seismology and time-series analysis for undergraduates.
- Coordinated multi-session lecture series, scheduling speakers and promoting student participation.

PEER-REVIEWED PUBLICATION

Cloud Seismology

- [C.2] Ni, Y., et al. (2025) **A Review of Cloud Computing and Storage in Seismology.** *Geophysical Journal International*, ggaf322. [https://doi.org/10.1093/gji/ggaf322]
- [C.1] Denolle, M., et al. (2025) **Training the Next Generation of Seismologists: Delivering Research-Grade Software Education for Cloud and HPC Computing through Diverse Training Modalities.** *Seismological Research Letters*. [https://doi.org/10.1785/0220240413]

Ambient Noise and Monitoring

- [A.4] Kidiwela, M., et al. (2026, accepted) **Active Protothrusts and Fluid Highways: Seismic Noise Reveals Hidden Subduction Dynamics in Cascadia** *Science Advances* [in press, https://arxiv.org/abs/2505.15731]
- [A.3] Feng, K.-F., et al. (2026) **A decadal survey of the near-surface seismic velocity response to hydrological variations in Utah, United States.** *Journal of Geophysical Research: Solid Earth*, 131(1), e2024JB030689. [https://doi.org/10.1029/2024JB030689]
- [A.2] Feng, K.-F., et al. (2021) **Controls on Seasonal Variations of Crustal Seismic Velocity in Taiwan Using Single-station Cross-component Analysis of Ambient Noise Interferometry.** *Journal of Geophysical Research: Solid Earth*, 126(11), e2021JB022650. [https://doi.org/10.1029/2021JB022650]
- [A.1] Feng, K.-F., et al. (2020) **Detecting Pre-eruptive Magmatic Processes of the 2018 Eruption at Kilauea, Hawaii Volcano with Ambient Noise Interferometry.** *Earth, Planets and Space*, 72, 74. [https://doi.org/10.1186/s40623-020-01199-x]

Earthquake Source and Seismogenic Structure

- [S.3] Hsu, Y.-F., et al. (2020) **Evidence for Fluid Migration During the 2016 Meinong Taiwan Aftershock Sequence.** *Journal of Geophysical Research: Solid Earth*, 125(9), e2020JB019994. [https://doi.org/10.1029/2020JB019994]
- [S.2] Lee, S.-J., et al. (2018) **Composite Megathrust Rupture from Deep Interplate to Trench of the 2016 Solomon Islands Earthquake.** *Geophysical Research Letters*, 45(2), 674-681. [https://doi.org/10.1002/2017GL076347]
- [S.1] Brown, D., et al. (2015) **Imaging High-pressure Rock Exhumation in Eastern Taiwan.** *Geology*, 43(7), 651-654. [https://doi.org/10.1130/G36810.1]